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**ROBUST CONTROL OF LINEAR SYSTEMS WITH
SWITCHED ACTUATORS SUBJECTED TO
DWEELL-TIME CONSTRAINTS**

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ROBUST CONTROL OF LINEAR SYSTEMS WITH SWITCHED ACTUATORS SUBJECTED TO DWELL-TIME CONSTRAINTS

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*"I am and always will be the optimist.
The hoper of far-flung hopes, and the dreamer of improbable dreams."*
— THE ELEVENTH DOCTOR

Resumo

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o sistema à trajetória cíclica alvo.

Abstract

Switched actuators are difficult to handle because of some characteristic constraints of their nature. but they are efficient in numerous engineering applications, being applied, for example, in attitude control systems for space vehicles. Thus, knowing how to use this type of actuator is important and necessary. The main problem to be solved in this thesis is the design of a controller that guarantees the robustness of a system with switched actuators subjected to time constraints that limit the response window of their commands. In this approach, a limit cycle target trajectory search is performed and then the error of the states compared to the target trajectory is calculated. Then a predictive control technique is applied to correct the error and get the system to the cyclic target trajectory

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List of Abbreviations and Acronyms

CTq	computed torque
DC	direct current
EAR	Equação Algébrica de Riccati
GDL	graus de liberdade
ISR	interrupção de serviço e rotina
LMI	linear matrices inequalities
MIMO	multiple input multiple output
PD	proporcional derivativo
PID	proporcional integrativo derivativo
PTP	point to point
UARMII	Underactuated Robot Manipulator II
VSC	variable structure control

List of Symbols

a	Distância
$\mathbf{a}$	Vetor de distâncias
$\mathbf{e}_j$	Vetor unitário de dimensão n e com o j -ésimo componente igual a 1
$\mathbf{K}$	Matriz de rigidez
m_1	Massa do cumpim
δ_{k-k_f}	Delta de Kronecker no instante k_f

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1 Introduction

1.1 Motivation

The dynamics of switched affine systems can be described by a state equation of the form $\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, t \in \mathbb{R}_+$. Where $x(t) \in \mathbb{R}^n$ is the state vector and $\sigma(t) \in \{1, 2, \dots, N\}$ is the switching state, which is associated to the matrix $A_{\sigma(t)} \in \mathbb{R}^{n \times n}$ and the vector $b_{\sigma(t)} \in \mathbb{R}^n$ at each time t .

The use of switched affine systems to solve problems in engineering can be seen in fields like biochemical reactions, where Parise *et al.* (2018) uses switched affine systems to represent the moments of evolution in a generic biochemical reaction network; urban traffic, where Hajiahmadi *et al.* (2016) approaches the urban traffic control; power converters, where Benmiloud *et al.* (2019) also uses this approach and other areas where the behavior of the system can be described as a switched affine system.

There is extensive research in this class of controllers, and most of them solve the problem by driving the system states to the neighborhood of the target states, not considering the trajectory behavior of the system. This can reduce the performance of the system when the system is affected by the trajectory of the limit cycle in steady operation.

This problem has been the target of many studies, Benmiloud *et al.* (2019) uses a hybrid Poincaré map approach to stabilize a switched affine system into the desired limit cycle and uses as an example a DC-DC converter. by Egidio *et al.* (2020b), which proposes a control design of a state-dependent switching function for discrete-time switched affine systems to drive the system to the desired limit cycle that is defined by design associated with the expected performance of the system. The solution was applied in an academic numerical example and a theoretical DC-DC converter. In Egidio *et al.* (2020a) the author expanded the work to continuous-time switched affine systems.

Some studies form the basis for the present work:

Vieira (2013) solved the problem for a particular class of affine systems. Here the controller drove the system to a target state using MILP (Mixed-Integer Linear Programming) and the cyclic trajectory was a consequence of the states as the system approached

the target.

Patino *et al.* (2010) defined the target trajectory using nonlinear programming and used this target trajectory in the predictive control in terms of errors.

Marcolino (2018) proposed a solution with a controller law that drives the switched affine system with dwell-time constraints to a target cyclic trajectory. In this work, the problem is addressed in two stages: first, the problem is interpreted as a convex optimization where the author can find a targeted cyclic trajectory that minimizes a cost function associated with the system performance; then a second part that uses the target trajectory as input and computes perturbation at the switching instants of the reference control signal using predictive control.

1.2 Contributions of the present work

This present work proposes to expand the solution from)matheus2018 and provide a broader solution that can be used for systems that have their dynamics changing over time.

In a system that can be represented by a switched affine system with the form

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}$$

there are cases in which the dynamic response of the system can change depending on the operation mode. This can be represented in switched affine systems with proper matrices $A_{\sigma(t)}$ and $b_{\sigma(t)}$ for each operation mode. With the current implementation of Marcolino (2018) considering an invariant system during its operation, there is a need for expanding the solution so it can represent more systems and solve more engineering problems that can be expressed as switched affine systems.

1.3 Outline of the remaining chapters

The remaining of this text is organized as follows. Chapter 2 describes the problem of finding a target periodic trajectory for the given switched affine system. Chapter 3 proposes then a predictive control law that drives the system to the target cyclic trajectory. Finally, chapter 4 brings some remarks and considerations of the work.

2 Determination of Target Periodic Trajectories

Devemos encontrar uma trajetória cíclica que respeite todos as restrições do sistema e minimize a distancia em torno do valor de referencia?

Função custo:

$$J(\tau^\infty, \Omega^\infty, s^\infty) = \min_{\tau, \Omega, s} \int_0^{T_p} [x(t) - \bar{x}^\infty]^T Q [x(t) - \bar{x}^\infty] dt \quad (2.1)$$

As entradas para a otimização são:

- Lista de modos de operação
- Lista de matriz A associada ao modo de operação
- Lista de matriz B associada ao modo de operação
- Matriz C
- Matriz D
- Lista de valores de controle em cada modo
- Valor de referência dos estados (xref)
- Tempo máximo de trajetória cíclica (Tpmax)
- Número máximo de estados
- Peso dos estados na função custo (Q)

Em que $Q = Q^T > 0$ e T_p é o período do ciclo.

A condição inicial é determinada de forma que o último ponto é coincidente com o primeiro ponto (dada a sequência de modos e instantes de chaveamento)

$$x(0) = (I - F_N F_{N-1} \dots F_1)^{-1} c \quad (2.2)$$

$$c = F_N F_{N-1} \dots F_2 G_1 u_1 + F_N F_{N-1} \dots F_3 G_2 u_2 + \dots + F_N G_{N-1} + G_N u_N \quad (2.3)$$

O resultado da otimizacao é τ^∞ e x_0

Exemplos:

Integrador (Matheus): Nao há otimizacao aqui, usamos a mesma trajetoria do Matheus para que tivessemos uma comparacao entre os resultados das simulacoes.

Patino, aqui estou usando um otimizador para encontrar a trajetoria de referencia.

Resultados:

Descrever resultados trajetorias Patino 1 e 2

TODO: FAZER SIMULACAO COMPLETA DE CADA CASO

2.1 Statement of the problem

2.2 Optimization

description of the optimization for finding the target trajectory

2.3 Calculating x_r

$$x_0 \in \mathfrak{R}$$

$$x(t) = e^{At} x(0) + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau \quad (2.4)$$

Modos:

$$\begin{aligned} [0 - t_1] : \dot{x} &= A_1 x + B_1 u_1 \\ [t_1 - t_2] : \dot{x} &= A_2 x + B_2 u_2 \\ &\vdots \\ [t_{N-1} - t_N] : \dot{x} &= A_N x + B_N u_N \end{aligned} \quad (2.5)$$

$$\begin{aligned}
x(t_1) &= e^{A_1 t_1} x(0) + \int_0^{t_1} e^{A_1(t_1-\tau)} B_1 u_1 d\tau \\
x(t_1) &= e^{A_1 t_1} x(0) + \left[\int_0^{t_1} e^{A_1(t_1-\tau)} B_1 d\tau \right] u_1 \\
x(t_1) &= F_1 x(0) + G_1 u_1
\end{aligned} \tag{2.6}$$

$$x(t_1) = F_1 x(0) + G_1 u_1 \tag{2.7}$$

$$x(t_2) = F_2 x(1) + G_2 u_2 \tag{2.8}$$

$$\vdots$$

$$x(t_N) = F_N x(N-1) + G_N u_N \tag{2.9}$$

$$\begin{aligned}
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N) \\
x(t_N) &= F_N F_{N-1} \dots F_1 x(0) + c
\end{aligned} \tag{2.10}$$

Queremos $x(t_N) = x(0) = ?$

$$\begin{aligned}
x(0) &= F_N F_{N-1} \dots F_1 x(0) + c \\
\underbrace{(I - F_N F_{N-1} \dots F_1)}_{\text{suposta invertível}} x(0) &= c \\
x(0) &= (I - F_N F_{N-1} \dots F_1)^{-1} c
\end{aligned} \tag{2.11}$$

em que

$$\begin{aligned}
c &= F_N F_{N-1} \dots F_2 G_1 x(1) + \\
&\quad F_N F_{N-1} \dots F_2 G_2 x(2) + \dots + F_N G_{N-1} x(N-1) + G_N x(N)
\end{aligned} \tag{2.12}$$

2.4 Application Examples

2.4.1 Variant Dynamic Circuit 1

2.4.2 Variant Dynamic Circuit 2

3 Predictive control: Preliminaries and notation

Consider a system with dynamics described by

$$x[k+1] = Ax[k] + Bu[k] \quad (3.1)$$

The objective is to drive the state $x[k]$ to the origin from a given initial condition while respecting state and control constraints in the form

$$x[k] \in \mathcal{X} \quad (3.2)$$

$$u[k] \in \mathcal{U} \quad (3.3)$$

$$\mathcal{U} = \{u \in \mathbb{R}^p : S_u u \leq b_u\} \quad (3.4)$$

and

$$\mathcal{X} = \{x \in \mathbb{R}^n : S_x x \leq b_x\} \quad (3.5)$$

3.1 Cost Function

The solution, if it exists, will not be unique. Therefore, it is necessary to adopt a criterion that allows for choosing the best sequence of control actions to perform the task.

The control problem consists of bringing the states to the origin, so we will employ a cost in the form

$$J = \left\| \hat{x}[k + N|k] \right\|_{P_f}^2 + \sum_{i=0}^{N-1} \left[\left\| \hat{x}[k + i|k] \right\|_Q^2 + \left\| \hat{u}[k + i|k] \right\|_R^2 \right] \quad (3.6)$$

and initial condition

$$\hat{x}[k|k] = x[k] \quad (3.7)$$

The expression for J consists of a **terminal cost** term and a trajectory **cost term**.

With P_f , Q , R being symmetric and positive-definite, with P_f chosen so that the terminal cost term $\left\| \hat{x}(k + N|k) - \bar{x} \right\|_{P_f}^2$ corresponds to the value function of the infinite horizon Discrete-Time Linear Quadratic Regulator (DLQR).

The cost function can be rewritten using the notation:

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN} \quad (3.8)$$

and then the sum equation

$$\begin{aligned} \sum_{j=0}^{N-1} \left\| \hat{u}[j+k|k] \right\|_R^2 &= \hat{u}^T[k|k] R \hat{u}[k|k] + \hat{u}^T[k+1|k] R \hat{u}[k+1|k] + \cdots \\ &\quad + \hat{u}^T[k+N-1|k] R \hat{u}[k+N-1|k] \end{aligned} \quad (3.9)$$

becomes

$$\underbrace{\begin{bmatrix} \hat{u}^T[k|k] & \hat{u}^T[k+1|k] & \cdots & \hat{u}^T[k+N-1|k] \end{bmatrix}}_{\hat{\mathbf{u}}^T} \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} \underbrace{\begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix}}_{\hat{\mathbf{u}}} \quad (3.10)$$

Using the notation

$$\mathbf{R} = \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} = \text{diag}_N(R) \quad (3.11)$$

we get to

$$\sum_{j=0}^{N-1} \left\| \hat{u}[j] \right\|_R^2 = \left\| \hat{\mathbf{u}} \right\|_{\mathbf{R}}^2 \quad (3.12)$$

Similarly, we can write

$$\begin{aligned} & \left\| \hat{x}[k+N|k] \right\|_{P_f}^2 + \sum_{j=0}^{N-1} \left\| \hat{x}[k+j|k] \right\|_Q^2 = x^T[k|k]Qx[k|k] \\ & + \hat{x}^T[k+1|k]Q\hat{x}[k+1|k] + \cdots + \hat{x}^T[k+N-1|k]Q\hat{x}[k+N-1|k] + \hat{x}^T[k+N|k]P_f\hat{x}[k+N|k] \\ & = \hat{x}^T[k|k]Q\hat{x}[k|k] + \\ & \underbrace{\begin{bmatrix} \hat{x}^T[k+1|k] & \hat{x}^T[k+2|k] & \cdots & \hat{x}^T[k+N|k] \end{bmatrix}}_{\hat{\mathbf{x}}[k]^T} \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} \underbrace{\begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix}}_{\hat{\mathbf{x}}[k]} \end{aligned} \quad (3.13)$$

with the notation

$$Q = \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} = \begin{bmatrix} \text{diag}_{N-1}(Q) & 0 \\ 0 & P_f \end{bmatrix} \in \mathbb{R}^{nN \times nN} \quad (3.14)$$

as

$$\left\| \hat{x}[k+N|k] \right\|_P^2 + \sum_{j=0}^{N-1} \left\| \hat{x}[k+j|k] \right\|_Q^2 = \left\| x[k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2 \quad (3.15)$$

So, the cost function can be written in the form

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \left\| x[k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2 + \left\| \hat{\mathbf{u}} \right\|_{\mathbf{R}}^2 \quad (3.16)$$

3.2 Restrictions

The restrictions are introduced aiming to drive the states of the system to the origin while solving

$$\min_{\hat{\mathbf{x}} \in \mathbb{R}^{nN}, \hat{\mathbf{u}} \in \mathbb{R}^{pN}} \mathbf{J}(\hat{\mathbf{x}}, \hat{\mathbf{u}}) \quad (3.17)$$

respecting constraints applied on

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathcal{U} \times \mathcal{U} \times \cdots \times \mathcal{U} = \mathcal{U}^N \quad (3.18)$$

$$\hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathcal{X} \times \mathcal{X} \times \cdots \times \mathcal{X}_f = \mathcal{X}^{N-1} \times \mathcal{X}_f \quad (3.19)$$

$\mathcal{X}_f$ is the maximum invariant and admissible set (MAS) with respect to the terminal state and control constraints under the action of a terminal control force.

$$\{\hat{x} \in \mathbb{R}^n : S_f \hat{x} \leq b_f\} \quad (3.20)$$

The constraint on $\hat{\mathbf{u}}$ can be written as

$$\begin{bmatrix} S_u & 0 & \cdots & 0 \\ 0 & S_u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_u \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \leq \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \end{bmatrix} \quad (3.21)$$

or, in a shorter form

$$\mathbf{S}_u \hat{\mathbf{u}} \leq \mathbf{b}_u \quad (3.22)$$

with

$$\mathbf{S}_u = \text{diag}_N(S_u), \mathbf{b}_u = \text{col}_N(b_u) \quad (3.23)$$

Similarly, the constraints on $\hat{\mathbf{x}}$ can be written as

$$\begin{bmatrix} S_x & 0 & \cdots & 0 \\ 0 & S_x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_x \end{bmatrix} \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \leq \begin{bmatrix} b_x \\ b_x \\ \vdots \\ b_x \end{bmatrix} \quad (3.24)$$

or, in short

$$\mathbf{S}_x \hat{\mathbf{x}} \leq \mathbf{b}_x \quad (3.25)$$

with

$$\mathbf{S}_x = \begin{bmatrix} \text{diag}_{N-1}(S_x) & 0 \\ 0 & S_f \end{bmatrix}, \mathbf{b}_x = \begin{bmatrix} \text{col}_{N-1}(b_x) \\ b_f \end{bmatrix} \quad (3.26)$$

3.3 Prediction Equation

Let's express the relationship between the vectors

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \quad (3.27)$$

based on the state equation

$$\hat{x}[k+i+1|k] = A\hat{x}[k|k] + B\hat{u}[k|k] \quad (3.28)$$

and initial condition $x[k]$.

The solution for the state equation is:

$$\hat{x}[k+i|k] = A^i \hat{x}[k|k] + \sum_{j=0}^{i-1} A^{i-1-j} B \hat{u}[k+j|k] \quad (3.29)$$

and it can be written for $k = 1, 2, \dots, N$ as

$$\hat{x}[k+1|k] = Ax[k] + B\hat{u}[k|k] \quad (3.30)$$

$$\hat{x}[k+2|k] = A^2x[k] + AB\hat{u}[k|k] + B\hat{u}[k+1|k] \quad (3.31)$$

$$\vdots \quad (3.32)$$

$$\hat{x}[k+N|k] = A^Nx[k] + A^{N-1}B\hat{u}[k|k] + \dots + AB\hat{u}[k+N-2|k] + B\hat{u}[k+N-1|k] \quad (3.33)$$

or

$$\begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} B & 0 & \dots & 0 \\ AB & B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \quad (3.34)$$

This can be written in a shorter form as

$$\hat{\mathbf{x}} = \mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \quad (3.35)$$

with

$$\mathbf{A} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \mathbf{B} = \begin{bmatrix} B & 0 & \dots & 0 \\ AB & B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \quad (3.36)$$

The relation

$$\hat{\mathbf{x}} = \mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \quad (3.37)$$

we can rewrite the cost function as

$$J = \left\| x[k|k] \right\|_P^2 + \overbrace{\left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2}^{\square} + \left\| \hat{\mathbf{u}} \right\|_{\mathbf{R}}^2 \quad (3.38)$$

$$\square = \left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2 \quad (3.39)$$

$$= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right]^T \mathbf{Q} \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right] \quad (3.40)$$

$$= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right]^T \mathbf{Q} \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \right] \quad (3.41)$$

$$= \left\| \mathbf{A}x[k|k] \right\|_{\mathbf{Q}}^2 + \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x[k|k] + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} \quad (3.42)$$

replacing $\square$ back in the cost equation and reordering the elements

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \left\| x[k|k] \right\|_P^2 + \left\| \mathbf{A}x[k|k] \right\|_{\mathbf{Q}}^2 \quad (3.43)$$

$$+ \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x[k|k] \quad (3.44)$$

$$+ \hat{\mathbf{u}}^T \mathbf{R} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} \quad (3.45)$$

The constant part will be denoted by c_0

Recalling that the matrix $\mathbf{Q}$ is symmetric, we have:

$$\left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} = \left\{ \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x[k|k] \right\}^T \quad (3.46)$$

As these terms are scalars, it can be concluded that they are equal to each other, therefore:

$$\begin{aligned} & \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + \hat{\mathbf{u}}^T \mathbf{B}^T \mathbf{Q} \mathbf{A} x(0) \\ &= 2 \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} \end{aligned} \quad (3.47)$$

Finally, it can be written

$$J(\hat{\mathbf{u}}) = \hat{\mathbf{u}}^T \left(\mathbf{R} + \mathbf{B}^T \mathbf{Q} \mathbf{B} \right) \hat{\mathbf{u}} + 2 \left[\mathbf{A}x[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}} + c_0 \quad (3.48)$$

with

$$c_0 = \left\| x(0) \right\|_P^2 + \left\| \mathbf{A}x(0) \right\|_{\mathbf{Q}}^2 \quad (3.49)$$

$$c_0 = \left\| x[k|k] \right\|_P^2 + \left\| \mathbf{A}x[k|k] \right\|_{\mathbf{Q}}^2$$

Note that we will now use the notation $J(\hat{\mathbf{u}})$ instead of $J(\hat{\mathbf{x}}, \hat{\mathbf{u}})$, since $\hat{\mathbf{x}}$ has been eliminated from the cost expression.

To complete the reformulation of the optimization problem in terms of $\hat{\mathbf{u}}$, the relation between $\hat{\mathbf{x}}$ and $\hat{\mathbf{u}}$ in the **restrictions** must be considered.

3.4 Optimization Problem

The optimization problem can be expressed as

$$\min_{\hat{\mathbf{x}} \in \mathbb{R}^{nN}, \hat{\mathbf{u}} \in \mathbb{R}^{pN}} J(\hat{x}, \hat{u}) \quad (3.50)$$

where

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \left\| x[k|k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_Q^2 + \left\| \hat{\mathbf{u}} \right\|_R^2 \quad (3.51)$$

with

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN}, \quad (3.52)$$

subject to

$$\mathbf{S}_x \hat{\mathbf{x}}[k] \leq \mathbf{b}_x \quad (3.53)$$

$$\mathbf{S}_u \hat{\mathbf{u}}[k] \leq \mathbf{b}_u \quad (3.54)$$

The vectors $\hat{\mathbf{x}}[k|k]$ and $\hat{\mathbf{u}}[k|k]$ are linked by the following **prediction equation**:

$$\hat{\mathbf{x}}[k+1|k] = \mathbf{A}\hat{x}[k|k] + \mathbf{B}\hat{\mathbf{u}}[k|k] \quad (3.55)$$

Thus, the cost can be rewritten as

$$\begin{aligned} J(\hat{\mathbf{u}}[k]) &= \hat{\mathbf{u}}[k]^T (\mathbf{R} + \mathbf{B}^T \mathbf{Q} \mathbf{B}) \hat{\mathbf{u}}[k] \\ &\quad + 2 \left[\mathbf{A} \hat{x}[k|k] \right]^T \mathbf{Q} \mathbf{B} \hat{\mathbf{u}}[k] + c_0 \end{aligned}$$

with

$$c_0 = \left\| \hat{x}[k|k] \right\|_P^2 + \left\| \mathbf{A} \hat{x}[k|k] \right\|_Q^2$$

and the constraints can be expressed as

$$\mathbf{S} \hat{\mathbf{u}}[k] \leq \mathbf{b} \quad (3.56)$$

with

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_x \mathbf{B} \\ \mathbf{S}_u \end{bmatrix}, \mathbf{b} = \begin{bmatrix} \mathbf{b}_x - \mathbf{S}_x \mathbf{A} \hat{x}[k|k] \\ \mathbf{b}_u \end{bmatrix} \quad (3.57)$$

Having obtained the solution $\hat{\mathbf{u}}[k|k]$ for the problem $\mathbb{P}(\hat{x}[k|k])$, the control to be applied to the plant is given by

$$\hat{u}[k|k] = \hat{u}^*[k|k]$$

where $\hat{u}^*[k|k]$ corresponds to the first p elements of $\hat{\mathbf{u}}^*[k|k]$.

3.5 Feasibility Region

slide: aula 8 slide 30

Pergunta: Para m dado N , como determinar o conjunto $\mathcal{D} \subset \mathcal{X}$ de condições iniciais $x(0)$ a partir das quais o conjunto alvo $\mathcal{X}_f$ pode ser atingido repetindo as restricoes sobre o estado e o controle ?

Alternativamente: Como determinar o conjunto $\mathcal{D}$ de condicoes iniciais $x(0)$ para as quais o problema de otimizacao e factivel

$\mathcal{D}$ é o conjunto (ou regioao) de pontos $x(0) \in \mathcal{X}$ para os quais exist $\mathbf{u} \in \mathbb{R}^{pN}$ que satisfaz as seguintes restricoes:

$$\begin{bmatrix} \mathbf{S}_x \mathbf{B} \\ \mathbf{S}_u \end{bmatrix} \mathbf{u} \leq \begin{bmatrix} \mathbf{b}_x - \mathbf{S}_x \mathbf{A} x(0) \\ \mathbf{b}_u \end{bmatrix} \quad (3.58)$$

Essas restricoes podem ser reescritas como

$$\begin{bmatrix} \mathbf{S}_x \mathbf{B} & \mathbf{S}_x \mathbf{A} \\ \mathbf{S}_u & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b}_x \\ \mathbf{b}_u \end{bmatrix} \quad (3.59)$$

ou ainda, acrescentando-se a restricao $S_x x(0) \leq b_x$,

$$\begin{bmatrix} \mathbf{S}_x \mathbf{B} & \mathbf{S}_x \mathbf{A} \\ \mathbf{S}_u & 0 \\ 0 & S_x \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b}_x \\ \mathbf{b}_u \\ b_x \end{bmatrix} \quad (3.60)$$

De forma resumida, pode-se escrever

$$S_w w \leq b_w \quad (3.61)$$

com

$$w = \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix}, S_w = \begin{bmatrix} S_x B & S_x A \\ S_u & 0 \\ 0 & S_x \end{bmatrix}, b_w = \begin{bmatrix} b_x \\ b_u \\ b_x \end{bmatrix} \quad (3.62)$$

A restric ao $S_w w \leq b_w$ define um poliedro $\mathcal{P}_w$ no espaco $\mathbb{R}^{(pN+n)}$ correspondente as variaveis $\mathbf{u}$, $x(0)$.

O conjunto $\mathcal{D}$   obtido **projetando-se** $\mathcal{P}_w$ sobre o subespaco de suas ultimas n componentes.

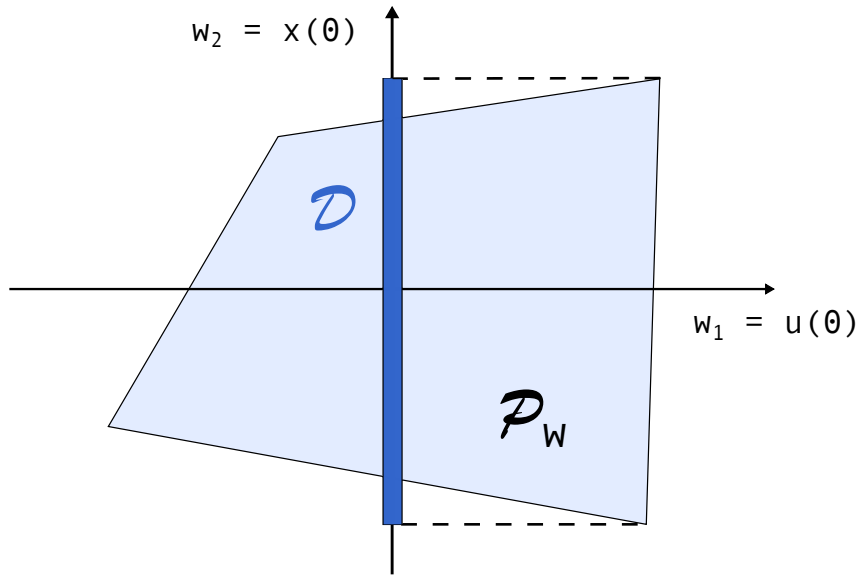


FIGURE 3.1 – Ilustrac o para $n = 1$ estado, $p = 1$ entrada e $N = 1$ passo

A **projec o** pode ser realizada empregando a funcao `projection` do MPT Toolbox:

$$Sw = [Sxn * Bn] \dots$$

4 Predictive Control for Stabilization of Periodic Trajectories

4.1 Statement of the problem

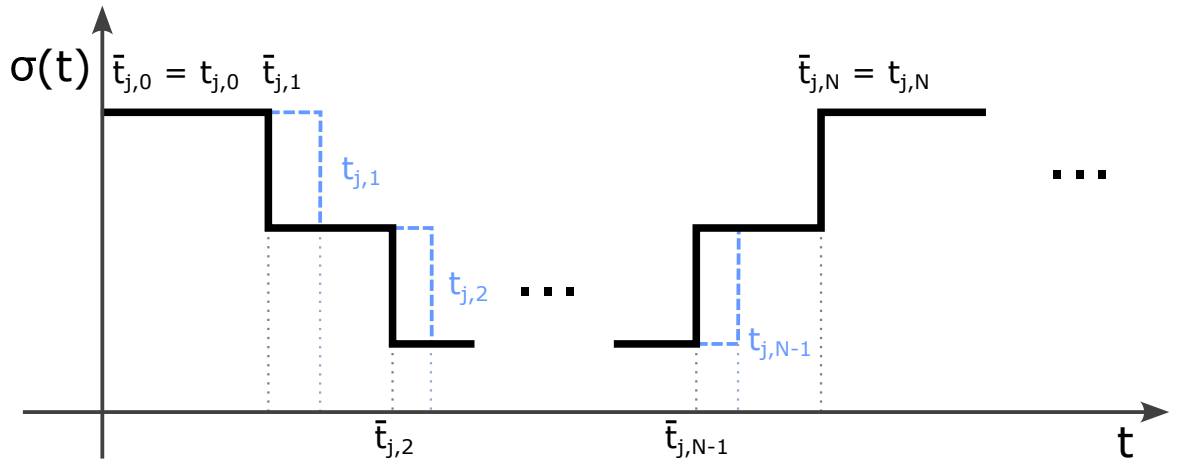


FIGURE 4.1 – Input signals

Considering a plant of an affine switched system described by:

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \quad (4.1)$$

The control action applied in the system is in the form of perturbations in each operation mode switching time. By allowing the control the a system using predictive control, we need to find a linear equation that relates the switching times and the resulting system at the end of one cycle.

4.2 Linearization procedure

The solution for the state equation (4.1) at time t_1 , starting from the initial condition $x(t_0)$, is of the form

$$x(t_1) = e^{A_0(t_1-t_0)}x(t_0) + \int_{t_0}^{t_1} e^{A_0(t_1-\tau)}B_0d\tau \quad (4.2)$$

$$\bar{x}(\bar{t}_1) = e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) + \int_{\bar{t}_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0d\tau \quad (4.3)$$

subtracting (4.3) from (4.2) we get

$$\begin{aligned} x(t_1) - \bar{x}(\bar{t}_1) &= \overbrace{e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0)}^{\square} \\ &\quad + \underbrace{\int_{t_0}^{t_1} e^{A_0(t_1-\tau)}B_0d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0d\tau}_{\bigcirc} \end{aligned} \quad (4.4)$$

expanding (4.4) first in $\square$:

$$\begin{aligned} \square &= e^{A_0(t_1-t_0)}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_0(\bar{t}_1+\delta t_1-t_0)}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_0(\bar{t}_1-t_0)}e^{\delta t_1}x(t_0) - e^{A_0(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= \bar{F}_0 e^{\delta t_1}x(t_0) - \bar{F}_0\bar{x}(\bar{t}_0) \\ &= \bar{F}_0(I + A_0\delta t_1)x(t_0) - \bar{F}_0\bar{x}(\bar{t}_0) \\ &= \bar{F}_0[x(t_0) - \bar{x}(\bar{t}_0)] + \bar{F}_0A_0\delta t_1 \end{aligned} \quad (4.5)$$

and then in $\bigcirc$:

$$\begin{aligned} \bigcirc &= \int_{t_0}^{t_1} e^{A_0(t_1-\tau)}B_0d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0d\tau \\ &= \int_{t_0}^{\bar{t}_1+\delta t_1} e^{A_0(\bar{t}_1+\delta t_1-\tau)}B_0d\tau - \int_{t_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0d\tau \\ &= e^{A_0\delta t_1} \left[\int_{t_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0d\tau + \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_0(\bar{t}_1-\tau)}B_0d\tau \right] - \int_{t_0}^{\bar{t}_1} e^{A_0(\bar{t}_1-\tau)}B_0d\tau \\ &= e^{A_0\delta t_1}[\bar{G}_0 + B_0\delta t_1] - \bar{G}_0 \\ &= (I + A_0\delta t_1)[\bar{G}_0 + B_0\delta t_1] - \bar{G}_0 \\ &= \cancel{\bar{G}_0} + B_0\delta t_1 + A_0\bar{G}_0\delta t_1 + \cancel{A_0B_0\delta t_1\delta t_1} - \cancel{\bar{G}_0} \\ &= A_0\bar{G}_0\delta t_1 + B_0\delta t_1 \end{aligned} \quad (4.6)$$

the resulting equations (4.5) and (4.6) are replaced back in (4.4) to complete the difference equation

$$\boxed{x(t_1) - \bar{x}(\bar{t}_1) = \bar{F}_0[x(t_0) - \bar{x}(\bar{t}_0)] + [\bar{F}_0A_0x(t_0) + A_0\bar{G}_0 + B_0]\delta t_1} \quad (4.7)$$

Repeating the steps for $x(t_2)$ we have

$$x(t_2) = e^{A_1(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau \quad (4.8)$$

$$\bar{x}(\bar{t}_2) = e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) + \int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau \int_a^b x^2 \phi \alpha dx \frac{1}{2} \quad (4.9)$$

subtracting (4.9) from (4.8)

$$\begin{aligned} x(t_2) - \bar{x}(\bar{t}_2) &= \overbrace{e^{A_1(t_2-t_1)}x(t_1) - e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1)}^{\square} \\ &\quad + \underbrace{\int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau}_{\bigcirc} \end{aligned} \quad (4.10)$$

expanding $\square$ in (4.10):

$$\begin{aligned} \square &= e^{A_1(\bar{t}_2+\delta t_2-\bar{t}_1-\delta t_1)}x(t_1) - e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\ &= e^{A_1(\bar{t}_2-\bar{t}_1)}e^{A_1(\delta t_2-\delta t_1)}x(t_1) - e^{A_1(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\ &= \bar{F}_1[I + A_1(\delta t_2 - \delta t_1)]x(t_1) - \bar{F}_1\bar{x}(\bar{t}_1) \\ &= \bar{F}_1[x(t_1) - \bar{x}(\bar{t}_1)] + \bar{F}_1 A_1 x(t_1) \delta t_2 - \bar{F}_1 A_1 x(t_1) \delta t_1 \end{aligned} \quad (4.11)$$

and expanding $\bigcirc$:

$$\begin{aligned}
 \bigcirc &= \int_{t_1}^{t_2} e^{A_1(t_2-\tau)} B_1 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau \\
 &= \int_{\bar{t}_1+\delta t_1}^{\bar{t}_2+\delta t_2} e^{A_1(\bar{t}_2+\delta t_2-\tau)} B_1 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau \\
 &= e^{A_1\delta t_2} \left[\int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau \right] \\
 &\quad - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau \\
 &= (I + A_1\delta t_2) \left[\bar{G}_1 - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_1(\bar{t}_2-\tau)} B_1 d\tau \right] - \bar{G}_1 \\
 &= (I + A_1\delta t_2) \left[\bar{G}_1 - e^{A_1(\bar{t}_2-\bar{t}_1)} B_1 \delta t_1 + B_1 \delta t_2 \right] - \bar{G}_1 \\
 &= (I + A_1\delta t_2) \left[\bar{G}_1 - \bar{F}_1 B_1 \delta t_1 + B_1 \delta t_2 \right] - \bar{G}_1 \\
 &= \cancel{\bar{G}_1} - \bar{F}_1 B_1 \delta t_1 + B_1 \delta t_2 + A_1 \bar{G}_1 \delta t_2 - \cancel{\bar{G}_1} \\
 &= A_1 \bar{G}_1 \delta t_2 + B_1 \delta t_2 - \bar{F}_1 B_1 \delta t_1
 \end{aligned} \tag{4.12}$$

and replacing (4.11) and (4.12) back into equation (4.10) we get to the difference equation for $x(t_2) - \bar{x}(\bar{t}_2)$

$$\begin{aligned}
 x(t_2) - \bar{x}(\bar{t}_2) &= \bar{F}_1 [x(t_1) - \bar{x}_1(\bar{t}_1)] \\
 &\quad - \bar{F}_1 [A_1 x(t_1) + B_1] \delta t_1 \\
 &\quad + [\bar{F}_1 A_1 \underbrace{x(t_1)}_{\triangle} + A_1 \bar{G}_1 + B_1] \delta t_2
 \end{aligned} \tag{4.13}$$

from (4.13), $\triangle$ can be expanded by replacing the resulting equation of (4.7)

$$\triangle = x(t_1) = \bar{x}(\bar{t}_1) + \bar{F}_0 [x(t_0) - \bar{x}(t_0)] + [\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 + B_0] \delta t_1 \tag{4.14}$$

note that when $\triangle$ is replaced in eq. (4.13) it will produce a product with δt_1 and δt_2 like

$$x(t_1) \delta t_1 = \bar{x}(\bar{t}_1) \delta t_1 + \bar{F}_0 \overbrace{[x(t_0) - \bar{x}(t_0)] \delta t_1}^{\square} + [\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 + B_0] \overbrace{\delta t_1 \delta t_1}^{\bigcirc}$$

both $\square$ and $\bigcirc$ are products of differences that produces a smaller number and they will be neglected in the first order approximations

$$x(t_1) \delta t_1 \doteq \bar{x}(\bar{t}_1) \delta t_1 \tag{4.15}$$

replacing (4.7) and (4.14) in (4.13) it follows that

$$\begin{aligned}
 x(t_2) - \bar{x}(\bar{t}_2) &= \bar{F}_1 \{ \bar{F}_0 [x(t_0) - \bar{x}(t_0) + [\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 + B_0] \delta t_1] \\
 &\quad - \bar{F}_1 [A_1 \bar{x}(t_1) + B_1] \delta t_1 \\
 &\quad + [\bar{F}_1 A_1 \bar{x}(t_1) + A_1 \bar{G}_1 + B_1] \delta t_2
 \end{aligned}$$

$$\begin{aligned}
 x(t_2) - \bar{x}(\bar{t}_2) &= \bar{F}_1 \bar{F}_0 [x(t_0) - \bar{x}(t_0)] \\
 &\quad + \bar{F}_1 [\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1] \delta t_1 \\
 &\quad + [\bar{F}_1 A_1 \bar{x}(\bar{t}_1) + A_1 \bar{G}_1 + B_1] \delta t_2
 \end{aligned} \tag{4.16}$$

Repeating the procedure for $x(t_3)$ we have

$$x(t_3) = e^{A_2(t_3-t_2)} x(t_2) + \int_{t_2}^{t_3} e^{A_2(t_3-\tau)} B_2 d\tau \tag{4.17}$$

$$\bar{x}(\bar{t}_3) = e^{A_2(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2) + \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \tag{4.18}$$

subtracting (4.17) from (4.18)

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) &= \overbrace{e^{A_2(t_3-t_2)} x(t_2) - e^{A_2(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2)}^{\square} \\
 &\quad + \underbrace{\int_{t_2}^{t_3} e^{A_2(t_3-\tau)} B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau}_{\bigcirc}
 \end{aligned} \tag{4.19}$$

expanding $\square$ in (4.19):

$$\begin{aligned}
 \square &= e^{A_2(\bar{t}_3+\delta t_3-\bar{t}_2-\delta t_2)} x(t_2) - e^{A_2(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2) \\
 &= e^{A_2(\bar{t}_3-\bar{t}_2)} e^{A_2(\delta t_3-\delta t_2)} x(t_2) - e^{A_2(\bar{t}_3-\bar{t}_2)} \bar{x}(\bar{t}_2) \\
 &= \bar{F}_2 [I + A_2(\delta t_3 - \delta t_2)] x(t_2) - \bar{F}_2 \bar{x}(\bar{t}_2) \\
 &= \bar{F}_2 [x(t_2) - \bar{x}(\bar{t}_2)] + \bar{F}_2 A_2 x(t_2) \delta t_3 - \bar{F}_2 A_2 x(t_2) \delta t_2
 \end{aligned} \tag{4.20}$$

and expanding $\bigcirc$:

$$\begin{aligned}
 \bigcirc &= \int_{t_2}^{t_3} e^{A_2(t_3-\tau)} B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \\
 &= \int_{\bar{t}_2+\delta t_2}^{\bar{t}_3+\delta t_3} e^{A_2(\bar{t}_3+\delta t_3-\tau)} B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \\
 &= e^{A_2\delta t_3} \left[\int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \right] \\
 &\quad - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \\
 &= (I + A_2\delta t_3) \left[\bar{G}_2 - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_2(\bar{t}_3-\tau)} B_2 d\tau \right] - \bar{G}_2 \\
 &= (I + A_2\delta t_3) \left[\bar{G}_2 - e^{A_2(\bar{t}_3-\bar{t}_2)} B_2 \delta t_2 + B_2 \delta t_3 \right] - \bar{G}_2 \\
 &= (I + A_2\delta t_3) \left[\bar{G}_2 - \bar{F}_2 B_2 \delta t_2 + B_2 \delta t_3 \right] - \bar{G}_2 \\
 &= \cancel{\bar{G}_2} - \bar{F}_2 B_2 \delta t_2 + B_2 \delta t_3 + A_2 \bar{G}_2 \delta t_3 - \cancel{\bar{G}_2} \\
 &= A_2 \bar{G}_2 \delta t_3 + B_2 \delta t_3 - \bar{F}_2 B_2 \delta t_2
 \end{aligned} \tag{4.21}$$

and then replacing (4.20) in (4.21) back in eq. (4.19) we get to the difference equation for $x(t_3) - \bar{x}(\bar{t}_3)$

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) &= \bar{F}_2 [x(t_2) - \bar{x}_2(\bar{t}_2)] \\
 &\quad - \bar{F}_2 [A_2 x(t_2) + B_2] \delta t_2 \\
 &\quad + [\bar{F}_2 A_2 x(t_2) + A_2 \bar{G}_2 + B_2] \delta t_3
 \end{aligned} \tag{4.22}$$

from (4.16), $\triangle$ can be expanded the same way as in the past iteration and $x(t_2) - \bar{x}(\bar{t}_2)$ can be replaced in (4.22)

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) &= \bar{F}_2 \{ \bar{F}_1 \bar{F}_0 [x(t_0) - \bar{x}(t_0)] \\
 &\quad + \bar{F}_1 [\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1] \delta t_1 \\
 &\quad + [\bar{F}_1 A_1 \bar{x}(t_1) + A_1 \bar{G}_1 + B_1] \delta t_2 \} \\
 &\quad - \bar{F}_2 [A_2 \bar{x}(t_2) + B_2] \delta t_2 \\
 &\quad + [\bar{F}_2 A_2 \bar{x}(t_2) + A_2 \bar{G}_2 + B_2] \delta t_3
 \end{aligned}$$

after rearranging the terms, one arrives at

$$\begin{aligned}
 x(t_3) - \bar{x}(\bar{t}_3) = & \bar{F}_2 \bar{F}_1 \bar{F}_0 [x(t_0) - \bar{x}(t_0)] \\
 & + \bar{F}_2 \bar{F}_1 [\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1] \delta t_1 \\
 & + \bar{F}_2 [\bar{F}_1 A_1 \bar{x}(t_1) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_2) + B_1 - B_2] \delta t_2 \\
 & + [\bar{F}_2 A_2 \bar{x}(t_2) + A_2 \bar{G}_2 + B_2] \delta t_3
 \end{aligned} \tag{4.23}$$

Now, there are enough iterations to deduce a pattern on each increment of the equation, so the equation for $x(t_i)$ can be deduced as

$$\begin{aligned}
 x(t_i) - \bar{x}(\bar{t}_i) = & \bar{F}_{i-1} \bar{F}_{i-2} \dots \bar{F}_0 [x(t_0) - \bar{x}(t_0)] \\
 & + \bar{F}_{i-1} \bar{F}_{i-2} \dots \bar{F}_1 [\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1] \delta t_1 \\
 & + \bar{F}_{i-1} \bar{F}_{i-2} \dots \bar{F}_2 [\bar{F}_1 A_1 x(t_1) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_2) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + \bar{F}_{i-1} [\bar{F}_{i-2} A_{i-2} \bar{x}(t_{i-2}) + A_{i-2} \bar{G}_{i-2} - A_{i-1} \bar{x}(\bar{t}_{i-1}) + B_{i-2} - B_{i-1}] \delta t_{i-1} \\
 & + [\bar{F}_{i-1} A_{i-1} \bar{x}(t_{i-1}) + A_{i-1} \bar{G}_{i-1} + B_{i-1}] \delta t_i
 \end{aligned}$$

(4.24)

and then, for $x(t_N)$ the equation is as follows

$$\begin{aligned}
 x(t_N) - \bar{x}(\bar{t}_N) = & \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_0 [x(t_0) - \bar{x}(t_0)] \\
 & + \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_1 \overbrace{[\bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1]}^{\square} \delta t_1 \\
 & + \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_2 [\bar{F}_1 A_1 x(t_1) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_2) + B_1 - B_2] \delta t_2 \\
 & \vdots \\
 & + \bar{F}_{N-1} [\bar{F}_{N-2} A_{N-2} \bar{x}(t_{N-2}) + A_{N-2} \bar{G}_{N-2} - A_{N-1} \bar{x}(\bar{t}_{N-1}) \\
 & \quad + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 & + \underbrace{[\bar{F}_{N-1} A_{N-1} \bar{x}(t_{N-1}) + A_{N-1} \bar{G}_{N-1} + B_{N-1}]}_{\circ} \delta t_N
 \end{aligned} \tag{4.25}$$

The equation (4.25) can be developed further to remove the dependencies on the

intermediary states $x(t_0), x(t_1), \dots, x(t_{N-2})$. By expanding $\square$ and $\bigcirc$, the following identity holds

$$\begin{aligned}
 \square &= \bar{F}_0 A_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
 &= A_0 \bar{F}_0 x(t_0) + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
 &= A_0 [\bar{x}(\bar{t}_1) - \bar{G}_0] + A_0 \bar{G}_0 - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
 &= A_0 \bar{x}(\bar{t}_1) - \cancel{A_0 \bar{G}_0} + \cancel{A_0 \bar{G}_0} - A_1 \bar{x}(\bar{t}_1) + B_0 - B_1 \\
 &= (A_0 - A_1) \bar{x}(\bar{t}_1) + B_0 - B_1
 \end{aligned} \tag{4.26}$$

$$\begin{aligned}
 \bigcirc &= \bar{F}_{N-1} A_{N-1} \bar{x}(t_{N-1}) + A_{N-1} \bar{G}_{N-1} + B_{N-1} \\
 &= A_{N-1} \bar{F}_{N-1} \bar{x}(t_{N-1}) + A_{N-1} \bar{G}_{N-1} + B_{N-1} \\
 &= A_{N-1} [\bar{x}(\bar{t}_N) - \bar{G}_{N-1}] + A_{N-1} \bar{G}_{N-1} + B_{N-1} \\
 &= A_{N-1} \bar{x}(\bar{t}_N) - \cancel{A_{N-1} \bar{G}_{N-1}} + \cancel{A_{N-1} \bar{G}_{N-1}} + B_{N-1} \\
 &= A_{N-1} \bar{x}(\bar{t}_N) + B_{N-1}
 \end{aligned} \tag{4.27}$$

replacing (4.26) and (4.27) in (4.25)

$$\begin{aligned}
 x(t_N) - \bar{x}(\bar{t}_N) &= \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_0 [x(t_0) - \bar{x}(t_0)] \\
 &\quad + \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_1 [(A_0 - A_1) \bar{x}(\bar{t}_1) + B_0 - B_1] \delta t_1 \\
 &\quad + \bar{F}_{N-1} \bar{F}_{N-2} \dots \bar{F}_2 [(A_1 - A_2) \bar{x}(\bar{t}_2) + B_1 - B_2] \delta t_2 \\
 &\quad \vdots \\
 &\quad + \bar{F}_{N-1} [(A_{N-2} - A_{N-1}) \bar{x}(\bar{t}_{N-1}) + B_{N-2} - B_{N-1}] \delta t_{N-1} \\
 &\quad + [A_{N-1} \bar{x}(\bar{t}_N) + B_{N-1}] \delta t_N
 \end{aligned}$$

(4.28)

and renaming the terms of (4.28) to

$$e(t) = x(t) - \bar{x}(\bar{t}) \quad (4.29)$$

$$\Phi = \bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_0 \quad (4.30)$$

$$\begin{aligned} \Gamma = & \begin{bmatrix} \bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_1[(A_0 - A_1)\bar{x}(\bar{t}_1) + B_0 - B_1], \\ \bar{F}_{N-1}\bar{F}_{N-2}\dots\bar{F}_2[(A_1 - A_2)\bar{x}(\bar{t}_2) + B_1 - B_2], \\ \vdots \\ \bar{F}_{N-1}[(A_{N-2} - A_{N-1})\bar{x}(\bar{t}_{N-1}) + B_{N-2} - B_{N-1}] \end{bmatrix} \end{aligned} \quad (4.31)$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \quad (4.32)$$

we get to the affine equation

$$e(t_N) = \Phi e(t_0) + \Gamma \delta t \quad (4.33)$$

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \quad (4.34)$$

where $\bar{x}$ is calculated as

$$\bar{x}(\bar{t}_1) = F_0\bar{x}(\bar{t}_0) + G_0 \quad (4.35)$$

$$\bar{x}(\bar{t}_2) = F_1\bar{x}(\bar{t}_1) + G_1 \quad (4.36)$$

$\vdots$

$$\bar{x}(\bar{t}_N) = F_{N-1}\bar{x}(\bar{t}_{N-1}) + G_{N-1} \quad (4.37)$$

and F_i and G_i are calculated as

$$[F_i, G_i] = c2dm(Ai, Bi, [], [], dti, 'zoh') \quad (4.38)$$

4.3 MPC Problem

Note that the equation (4.33) relates to the equation (3.1) as both describe the evolution of a system over time. The equation (3.1) represents the system's state and input at a given time step, and the equation (4.33) describes the error between the system's predicted and actual values at two different time points.

Referring to equation (3.1), we can make some substitutions to simplify our discussion. We'll replace variable x with e , and u with δt . Additionally, we'll denote the cycle index with k . With these substitutions in place, we arrive at equation (4.33).

4.3.1 Cost Function

$$J = \left\| e[N_p] \right\|_{P_f}^2 + \sum_{j=0}^{N_p-1} \left\| e[j] \right\|_Q^2 + \left\| \delta t[j] \right\|_R^2 \quad (4.39)$$

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \quad (4.40)$$

where $Q = Q^T > 0$, $\rho > 0$

$$\delta t = [t_{j,1} - \bar{t}_{j,1}, t_{j,2} - \bar{t}_{j,2}, \dots]^T \quad (4.41)$$

4.3.2 Dwell-time Constraints

We want to enforce a constraint on the dwell times where the switching times will be subject to the constraints:

$$(t_i - t_{i-1}) \geq \Delta t_{i,min}, \quad i = 1, 2, \dots, N \quad (4.42)$$

then we can write the restriction as

$$\delta t_1 \geq -\Delta \bar{t}_1 + \Delta t_{1,min} \quad (4.43)$$

$$\delta t_i - \delta t_{i-1} \geq -\Delta \bar{t}_i + \Delta t_{i,min}, \quad i = 1, 2, \dots, N \quad (4.44)$$

$$\delta t_{N-1} \geq -\Delta \bar{t}_N + \Delta t_{N,min} \quad (4.45)$$

$$\Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1} \quad (4.46)$$

to match the constraint from (3.56)

$$\mathbf{S}\hat{\mathbf{u}}[k] \leq b$$

the equations can be written as

$$-\delta t_1 \leq +\Delta \bar{t}_1 - \Delta t_{1,min} \quad (4.47)$$

$$-\delta t_i + \delta t_{i-1} \leq +\Delta \bar{t}_i - \Delta t_{i,min} , \quad i = 1, 2, \dots, N \quad (4.48)$$

$$-\delta t_{N-1} \leq +\Delta \bar{t}_N - \Delta t_{N,min} \quad (4.49)$$

with

$$\mathbf{L} = \begin{bmatrix} -1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & -1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \quad (4.50)$$

$$\mathbf{c} = \begin{bmatrix} \Delta t_{r_1} - \Delta t_{r_1,min} \\ \Delta t_{r_2} - \Delta t_{r_2,min} \\ \vdots \\ \Delta t_{r_{N-1}} - \Delta t_{r_{N-1},min} \\ \Delta t_{r_N} - \Delta t_{r_N,min} \end{bmatrix} \quad (4.51)$$

4.3.3 Prediction Equation

Prediction Equation placeholder

4.3.4 Optimization Problem

Optimization Problem placeholder

4.3.5 Feasibility Region

Feasibility Region placeholder

4.4 Application Examples

4.4.1 Integrador duplo

4.4.2 Patino 1

4.4.3 Patino 2

Matrizes do artigo patino exemplo 2:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & -\frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \quad (4.52)$$

$$A = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \quad (4.53)$$

$$\begin{aligned} \mathbf{u}_0 &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T \\ \mathbf{u}_1 &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T \\ \mathbf{u}_2 &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \\ \mathbf{u}_3 &= \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}^T \\ \mathbf{u}_4 &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \\ \mathbf{u}_5 &= \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^T \\ \mathbf{u}_6 &= \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T \\ \mathbf{u}_7 &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T \end{aligned} \quad (4.54)$$

$$A_{u_0} = A\mathbf{u}_0 = \begin{bmatrix} 0 \\ 0 \\ -\frac{Rx_3(t)}{L} \end{bmatrix} \quad (4.55)$$

$$A_{u_1} = A\mathbf{u}_1 = \begin{bmatrix} 0 \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \quad (4.56)$$

$$A_{u_2} = A\mathbf{u}_2 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ -\frac{x_3(t)}{C_2} \\ \frac{x_2(t)-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \quad (4.57)$$

$$A_{u_3} = A\mathbf{u}_3 = \begin{bmatrix} \frac{x_3(t)}{C_1} \\ 0 \\ \frac{E-x_1(t)-Rx_3(t)}{L} \end{bmatrix} \quad (4.58)$$

$$A_{u_4} = A\mathbf{u}_4 = \begin{bmatrix} -\frac{x_3(t)}{C_1} \\ 0 \\ \frac{x_1(t)-Rx_3(t)}{L} \end{bmatrix} \quad (4.59)$$

$$A_{u_5} = A\mathbf{u}_5 = \begin{bmatrix} -\frac{x_3(t)}{C_1} \\ \frac{x_3(t)}{C_2} \\ \frac{E-x_1(t)-x_2(t)-Rx_3(t)}{L} \end{bmatrix} \quad (4.60)$$

$$A_{u_6} = A\mathbf{u}_6 = \begin{bmatrix} 0 \\ -\frac{x_3(t)}{C_2} \\ \frac{x_2(t)-Rx_3(t)}{L} \end{bmatrix} \quad (4.61)$$

$$A_{u_7} = A\mathbf{u}_7 = \begin{bmatrix} 0 \\ 0 \\ \frac{E-Rx_3(t)}{L} \end{bmatrix} \quad (4.62)$$

$$\dot{X} = A_{u_0}X + B_0u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (4.63)$$

$$\dot{X} = A_{u_1}X + B_1u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (4.64)$$

$$\dot{X} = A_{u_2}X + B_2u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & -\frac{1}{C_2} \\ -\frac{1}{L} & \frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (4.65)$$

$$\dot{X} = A_{u3}X + B_3u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (4.66)$$

$$\dot{X} = A_{u4}X + B_4u = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ -\frac{1}{L} & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (4.67)$$

$$\dot{X} = A_{u5}X + B_5u = \begin{bmatrix} 0 & 0 & -\frac{1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (4.68)$$

$$\dot{X} = A_{u6}X + B_6u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_2} \\ 0 & -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u \quad (4.69)$$

$$\dot{X} = A_{u7}X + B_7u = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} u \quad (4.70)$$

4.5 Conclusion

mostrar o modelo em malha aberta indo para a sua trajetoria ciclica estavel junto com o modelo com controle ativo mostrando a convergencia mais rapida

5 Conclusão

5.1 RASCUNHO

- > artigo mateus: sessao 4 (38)
- > usar como referencia para escrever as restricoes de dwell time
- > usando a restricao da 3.4, $S_u = -L$, $B_u = -c$

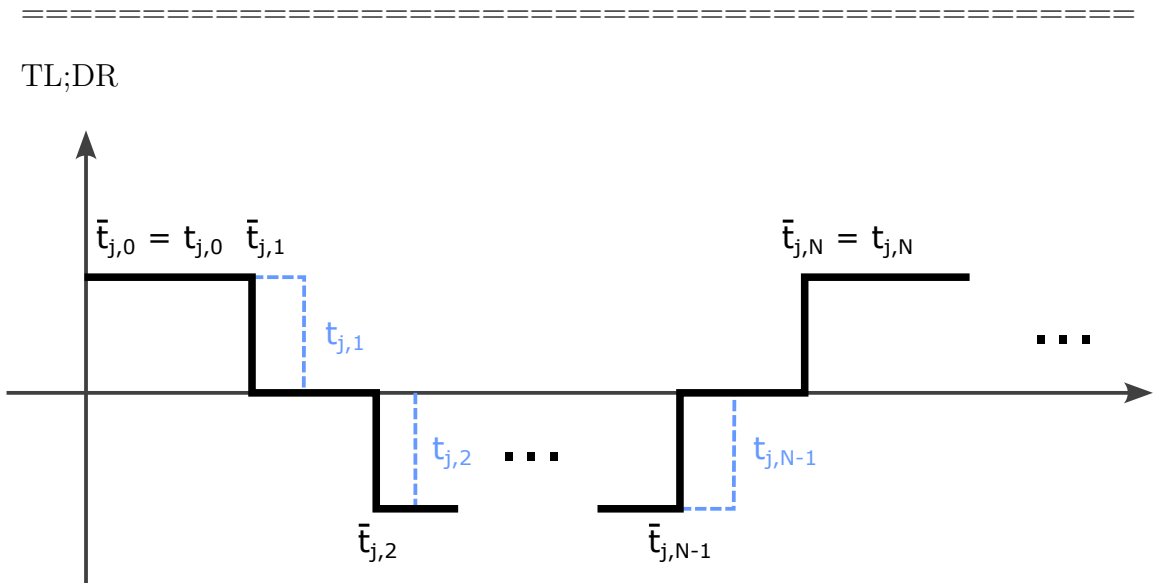


FIGURE 5.1 – Instantes de chaveamento

a restricao que queremos impor é:

$$(t_i - t_{i-1}) \geq \Delta t_{i,min} , \quad i = 1, 2, \dots, N \quad (5.1)$$

como

$$\Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1} \quad (5.2)$$

então podemos escrever a restricao da seguinte maneira

$$\delta t_1 \geq -\Delta \bar{t}_1 + \Delta t_{1,min} \quad (5.3)$$

$$\delta t_i - \delta t_{i-1} \geq -\Delta \bar{t}_i + \Delta t_{i,min} , \quad i = 1, 2, \dots, N \quad (5.4)$$

$$\delta t_{N-1} \geq -\Delta \bar{t}_N + \Delta t_{N,min} \quad (5.5)$$

que pode ser representado na forma matricial como

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \\ 0 & 0 & 0 & \cdots & 0 & -1 \end{bmatrix} \quad (5.6)$$

$$c = \begin{bmatrix} -\Delta t_{r_1} + -\Delta t_{r_1,min} \\ -\Delta t_{r_2} + -\Delta t_{r_2,min} \\ \vdots \\ -\Delta t_{r_{N-1}} + -\Delta t_{r_{N-1},min} \\ -\Delta t_{r_N} + -\Delta t_{r_N,min} \end{bmatrix} \quad (5.7)$$

=====

4 Predictive Control Formulation

The problem addressed herein consists in determining suitable perturbations in the switching times of the input signal to drive the state deviation $e[k]$ to zero asymptotically. As a result, the state $x[k]$ will be driven to the target point $x_r(t_0) = x_r(t_N)$.

For this purpose, the discrete-time model (28) derived in the previous section is employed within a predictive control framework. The cost function to be minimized at the beginning of the k th oscillation cycle is chosen as

$$J = \left\| \hat{e} \left[k = N_p[k] \right] \right\|_{P_f}^2 + \sum_{j=1}^{N_p} \left\| \hat{e} \left[k + j|k \right] \right\|_Q^2 + \left\| \hat{\delta}_t \left[k + j - 1|k \right] \right\|_R^2 \quad (5.8)$$

where $Q = Q_T \in \mathbb{R}^{n \times n}$, $R = R_T \in \mathbb{R}^{(N-1) \times (N-1)}$, $P_f = P_f^T \in \mathbb{R}^{n \times n}$ are positive-definite weight matrices. Following standard predictive control notation, the hat and $[k]$ symbols are used to indicate that $\hat{\delta}t[k|k]$, $\hat{\delta}t[k+1|k]$, $\hat{\delta}t[k+N_p|k]$ is a sequence of manipulated vectors which are to be optimized over a prediction horizon spanning N_p oscillation cycles. The relation between these manipulated vectors and the predicted state deviations is given by.

$$\begin{aligned}
\hat{e}[k|k] &= e[k] \\
\hat{e}[k+j|k] &= \Phi \hat{e}[k+j-1|k] + \Gamma \hat{\delta}t[k+j-1|k], \\
j &= 1, 2, \dots, N_p
\end{aligned} \tag{5.9}$$

In order to enforce (8) - (10) at each oscillation cycle, the following constraints will be imposed:

$$\mathbf{L} \hat{\delta}t[k+j-1|k] \geq c, j = 1, 2, \dots, N_p \tag{5.10}$$

with $\mathbf{L} \in \mathbb{R}^{N \times (N-1)}$ and $c \in \mathbb{R}^{N \times (N-1)}$ and $c \in \mathbb{R}^N$ given by

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \\ 0 & 0 & 0 & \cdots & 0 & -1 \end{bmatrix} \tag{5.11}$$

$$c = \begin{bmatrix} -\Delta t_{r_1} + -\Delta t_{r_1, min} \\ -\Delta t_{r_2} + -\Delta t_{r_2, min} \\ \vdots \\ -\Delta t_{r_{N-1}} + -\Delta t_{r_{N-1}, min} \\ -\Delta t_{r_N} + -\Delta t_{r_N, min} \end{bmatrix} \tag{5.12}$$

Moreover, nominal stability for the closed-loop system can be obtained by imposing a constraint on the form

$$\hat{e}[k+N_p|k] \in \mathbb{E}_f \tag{5.13}$$

Bibliography

BENMILOUD, M.; BENALIA, A.; DJEMAI, M.; DEFOORT, M. On the local stabilization of hybrid limit cycles in switched affine systems. **IEEE Transactions on Automatic Control**, v. 64, n. 2, p. 841–846, 2019.

EGIDIO, L.; DEAECTO, G.; GEROMEL, J. Limit cycle global asymptotic stability of continuous-time switched affine systems. **IFAC-PapersOnLine**, v. 53, p. 6121–6126, 01 2020.

EGIDIO, L. N.; DAIHA, H. R.; DEAECTO, G. S. Global asymptotic stability of limit cycle and h_2/h_∞ performance of discrete-time switched affine systems. **Automatica**, v. 116, p. 108927, 2020. ISSN 0005-1098.

HAJIAHMADI, M.; SCHUTTER, B. D.; HELLENDORRN, H. Robust h_∞ switching control techniques for switched nonlinear systems with application to urban traffic control. **International Journal of Robust and Nonlinear Control**, Wiley Online Library, v. 26, n. 6, p. 1286–1306, 2016.

MARCOLINO, M. H. **Stabilization of Periodic trajectories through receding horizon optimization of switching times**. 2018. 79 f. Tese (Doutorado em Sistemas de Controle) — Instituto Tecnológico de Aeronáutica, São José dos Campos, 2018.

PARISE, F.; VALCHER, M. E.; LYGEROS, J. Computing the projected reachable set of stochastic biochemical reaction networks modeled by switched affine systems. **IEEE Transactions on Automatic Control**, v. 63, n. 11, p. 3719–3734, 2018.

PATINO, D.; RIEDINGER, P.; RUIZ, F. A predictive control approach for dc-dc power converters and cyclic switched systems. In: IEEE. **2010 IEEE International Conference on Industrial Technology**. [S.l.], 2010. p. 1259–1264.

VIEIRA, M. S. **Controle Preditivo de sistemas com atuadores sujeitos a restrições temporais de chaveamento**. 2013. 119 f. Dissertação (Mestrado em Sistemas de Controle) — Instituto Tecnológico de Aeronáutica, São José dos Campos, 2013.

Appendix A - Tópicos de Dilema Linear

A.1 Modelo de simulação

```
+engine
├─ CORES.m
├─ fun_custo_patino.m
├─ get_config_sim_matheus.m
├─ get_config_sim_patino_1.m
├─ get_config_sim_patino_2.m
├─ get_config.m
├─ get_otmin_opt.m
├─ get_ts.m
├─ get_x0.m
├─ get_xr.m
├─ nonlincon_patino.m
├─ otmin.m
├─ sim_1.m
├─ sim_n.m
└─ +mpc
    ├─ construcao_modelo_instantes.m
    ├─ matrizes_ss_mpc_dualmode_switching.m
    └─ mpc_dualmode_switching.m
```

A.1.1 get_x0

funcao de calculo da condicao inicial

A.1.2 get_xf

Texto do modelo de simulação sendo colocado aqui.

Annex A - Exemplo de um Primeiro Anexo

A.1 Uma Seção do Primeiro Anexo

Algum texto na primeira seção do primeiro anexo.

FOLHA DE REGISTRO DO DOCUMENTO

1. CLASSIFICAÇÃO/TIPO TD	2. DATA 25 de março de 2015	3. DOCUMENTO Nº DCTA/ITA/DM-018/2015	4. Nº DE PÁGINAS 50
5. TÍTULO E SUBTÍTULO: Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints			
6. AUTOR(ES): Daniel Vieira			
7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES): Instituto Tecnológico de Aeronáutica – ITA			
8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR: Cupim; Cimento; Estruturas			
9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO: Cupim; Dilema; Construção			
10. APRESENTAÇÃO: (X) Nacional () Internacional ITA, São José dos Campos. Curso de Mestrado. Programa de Pós-Graduação em Engenharia Aeronáutica e Mecânica. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Adalberto Santos Dupont. Coorientadora: Prof ^a . Dr ^a . Doralice Serra. Defesa em 05/03/2015. Publicada em 25/03/2015.			
11. RESUMO: Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.			
12. GRAU DE SIGILO: (X) OSTENSIVO () RESERVADO () SECRETO			